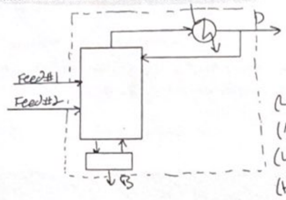
10

RAYAN BAIG
CHE 313
HOMEWORK #5

1) PROBLEM 5.D5

D5. A distillation column with a total condenser and a partial reboiler is separating two feeds. The first feed is a saturated liquid, and its rate is 100.0 kmol/h. This feed is 55.0 mol% methanol, 21.0 mol% ethanol, 23.0 mol% propanol, and 1.0 mol% butanol. The second feed is a saturated liquid with a flow rate of 150.0 kmol/h. This feed is 1.0 mol% methanol, 3.0 mol% ethanol, 26.0 mol% propanol, and 70.0 mol% butanol. We want to recover 99.3% of the propanol in distillate and 99.5% of the butanol in bottoms. Make appropriate assumptions, and find distillate and bottoms flow rates and mole fractions of distillate and bottoms.

Assumptions:
* All methanol & ethanol goes to the distillate



	Feed#1	Feed#2
	$F_1 = 100$ kmol/h	$F_2 = 150$ kmol/h
(LNK) Methanol:	55%	1%
(NK) Ethanol:	21%	3%
(LK) Propanol:	23%	26%
(HK) Butanol	1%	70%

Want 99.3% Propanol in Distillate & 99.5% Butanol in Bottoms

$D x_{methanol,D} = (100 \frac{kmol}{h})(0.55) + (150 \frac{kmol}{h})(0.01)$

$* \; D x_{methanol,D} = 56.5$ kmol/h

$D x_{Ethanol,D} = (100)(0.21) + (150)(0.03)$

$* \; D x_{Ethanol,D} = 25.5$ kmol/h

$D x_{Propanol,D} = (0.993)[(100)(0.23) + (150)(0.26)]$

$* \; D x_{Propanol,D} = 61.566$ kmol/h

$D x_{Butanol,D} = (1-0.995)[(100)(0.01) + (150)(0.70)]$

$* \; D x_{Butanol,D} = 0.53$ kmol/h

$\sum D x_{i,D} = D = 144.096$ kmol/h

$F_1 + F_2 = D + B$

$B = F_1 + F_2 - D = 100 + 150 - 144.096$

$B = 105.904$ kmol/h

CHECK

$B x_{Butanol,B} = (0.995)[(100)(0.01) + (150)(0.70)]$

$* \; B x_{Butanol,B} = 105.47$ kmol/h

$B x_{Propanol,B} = (1-0.993)[(100)(0.23) + (150)(0.26)]$

$* \; B x_{Propanol,B} = 0.434$ kmol/h

$\sum B x_{i,B} = B = 105.904$ kmol/h ✓

Calculating Mole frac = $\frac{D x_i}{D}$ ... or B...

$$\boxed{\begin{aligned} D &= 144.096 \text{ kmol/h} \\ B &= 105.904 \text{ kmol/h} \\ x_{methanol,D} &= 0.392 \\ x_{Ethanol,D} &= 0.1770 \\ x_{propanol,D} &= 0.4273 \\ x_{Butanol,D} &= 0.0037 \\ x_{methanol,B} &= 0 \\ x_{Ethanol,B} &= 0 \\ x_{propanol,B} &= 0.0041 \\ x_{Butanol,B} &= 0.9959 \end{aligned}}$$

Check mole frac sums

$\sum x_D = 1.00$ ✓

$\sum x_B = 1.00$ ✓

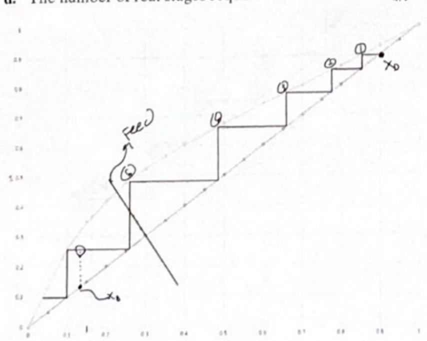
2) PROBLEM 4.D8

D8.* For Problem 4.D7c for separation of acetone from ethanol, determine

a. The number of stages required at total reflux.
b. The values of $(L/V)_{min}$ and $(L/D)_{min}$.
c. How much larger the L/D used is than $(L/D)_{min}$.
d. The number of real stages required for L/V = 0.8 if $E_{MV} = 0.75$.

FROM HW #4 PROBLEM 4D7 $q = 0.675$

A)



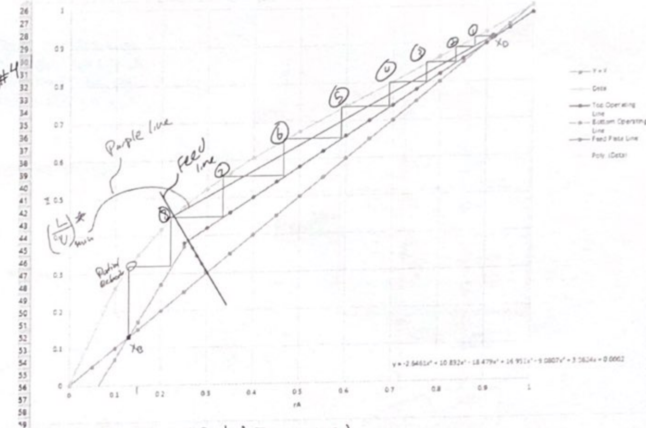
When there is total reflux op lines approach the y=x line.

At total reflux $\approx 5 \; \& \; \frac{3}{4} = \boxed{5.75 \text{ stages required}}$

B) $\left(\frac{L}{V}\right)_{min}$ & $\left(\frac{L}{D}\right)_{min}$

Finding where feed line intersects w/ equilibrium data

FROM MY HW#4



Feed Line

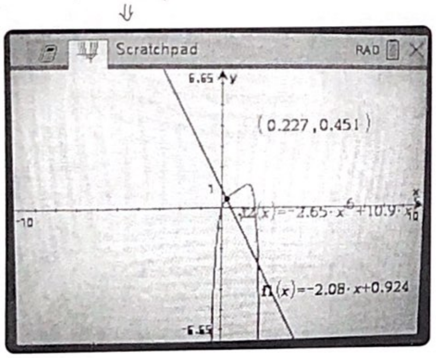
$\hookrightarrow y = -2.08x + 0.924$

6th order polynomial

$\hookrightarrow y = -2.6461x^6 + 10.892x^5 - 18.479x^4 + 16.951x^3 - 9.0807x^2 + 3.3624x + 0.0002$

Using Graphing Calculator

⇓



→ intersects @ (0.227, 0.451)

$\left(\frac{L}{V}\right)_{min} = \frac{x_D - y^*}{x_D - x^*} = \frac{0.9 - 0.451}{0.9 - 0.227} = 0.667$

$\left(\frac{L}{D}\right)_{min} = \frac{\left(\frac{L}{V}\right)_{min}}{1 - \left(\frac{L}{V}\right)_{min}} = 2.004$

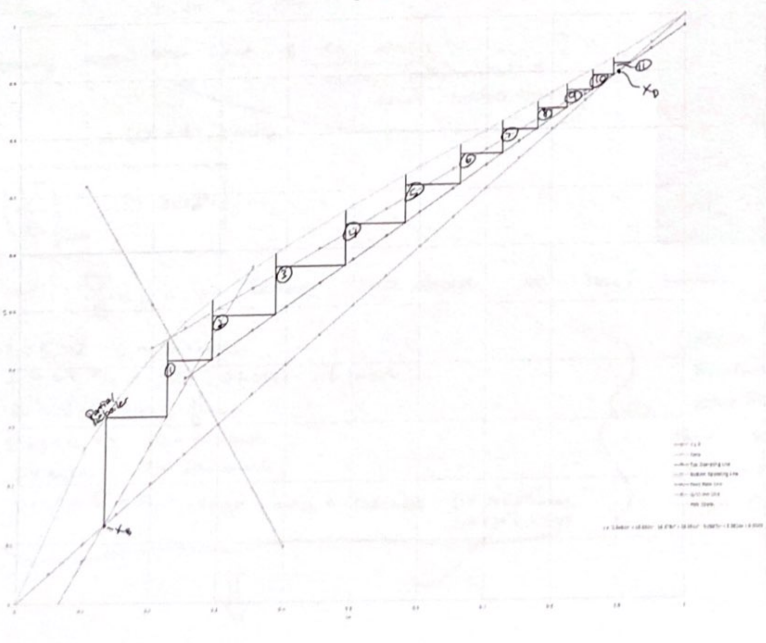
$$\boxed{\left(\frac{L}{V}\right)_{min} = 0.667 \quad \left(\frac{L}{D}\right)_{min} = 2.004}$$

C) $\frac{L}{D_{actual}} = \frac{\frac{L}{V}}{1 - \frac{L}{V}} = \frac{0.8}{1 - 0.8} = \underbrace{4}_{actual}$ (FROM 4.D.7) $\left(\frac{L}{D}\right)_{min} = 2.004$ ∴

ratio of how much larger $\frac{L}{D}$ is than $\frac{L}{D\,min}$ $= \frac{\left(\frac{L}{D}\right)}{\left(\frac{L}{D_{m}}\right)} = 1.996$ (I avoided rounding)

$\boxed{\text{1.996 times larger}}$

D)



Finding eq. for $\left(\frac{L}{D}\right)_{min}$

$y = 0.667x + b$

Line w/ (0.9, 0.9)

$y = 0.667x + 0.2997$

* OP. Lines Are the Same as Before

$E_{MV} = 0.75$ ∴ 0.75

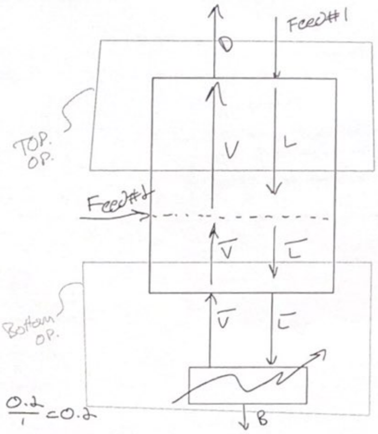
11 REAL STAGES & A PARTIAL REBOILER OPTIMUM FEED STAGE ⇒ STAGE #1

3) PROBLEM 4.D18

D18. A stripping column with two feeds is separating acetone and ethanol at 1.0 atm. Feed F_1 is a saturated liquid and is fed into the top of column (no condenser). Flow rate of F_1 is 100.0 kmol/h, and this feed is 60.0 mol% acetone. Feed F_2 is 40.0 mol% acetone, it is a two-phase feed that is 80% vapor, and flow rate is 80.0 kmol/h. Bottoms is 4.0 mol% acetone. The column has a partial reboiler.

a. Calculate $(\bar{V}/B)_{min}$.

b. If $(\bar{V}/B) = 1.5$, find D and y_D, the optimum feed stage for feed F_2, and the total number of stages. Step off stages from the bottom upward.



FEED#1

$F_1 = 100$ kmol/h

$z_{acetone} = 0.60$

* Sat liquid

⇓

Feed#1 Line is $q = \infty$

* $x = 0.60$

FEED#2

$F_2 = 80.0$ kmol/h

$z_2 = 0.40$

* Two phase

80% vapor

20% liquid

Feed #2 Line $q_2 = \frac{L}{F} = \frac{0.2}{1} = 0.2$

$y = \frac{q_2}{q_2-1}x - \frac{z_2}{q_2-1}$

$y = \frac{\frac{2}{10}}{\frac{2}{10}-\frac{10}{10}}x - \frac{0.40}{\frac{2}{10}-1}$

$y = \frac{\frac{2}{10}}{\frac{-8}{10}}x + \frac{\frac{4}{10}}{\frac{8}{10}}$

* $y = \frac{-1}{4}x + \frac{1}{2}$

Bottom op intersects w/ $x_B = 0.04$ on $y=x$ & the feed#2 Line

CMO Analysis

$L = F_1$

$\bar{L} = L + F_{2L}$

MB: $\{ \bar{L} = \bar{V} + B$

$V = \bar{V} + F_{2V}$

$\bar{V} = F_{2V}$

A) $\left(\frac{\bar{V}}{B}\right)_{min} = \frac{\bar{V}}{\bar{L}-\bar{V}} = \frac{1}{\left(\frac{\bar{L}}{\bar{V}}\right)_{max} - 1} = 0.808$

Find $\left(\frac{\bar{L}}{\bar{V}}\right)_{max}$

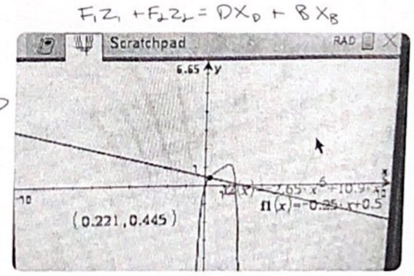
$\left(\frac{\bar{L}}{\bar{V}}\right)_{max} = \frac{0.445 - 0.04}{0.221 - 0.04} = 2.24$

Using Feed #2 Line & eq. data

make polynomial & find intersection w/

$= (0.221, 0.445)$

⇒



Overall MB & CB

$F_1 + F_2 = D + B$

$F_1 z_1 + F_2 z_2 = D x_D + B x_B$

$\boxed{\left(\frac{\bar{V}}{B}\right)_{MAX} = 0.808}$

B) w/ $\frac{\bar{V}}{B} = 1.5$ ⇒ using CMO above we now know:

$L = F_1 \Rightarrow L = 100$ kmol/h

$\bar{L} = L + F_{2L} = 100 + 0.2(F_2) = 116$ kmol/h

$\bar{L} = \bar{V} + B$ using $\frac{\bar{V}}{B} = 1.5$

$\bar{L} = 2.5B \Rightarrow B = 46.4$ kmol/h

$\bar{V} = B(1.5) \Rightarrow \bar{V} = 69.6$ kmol/h

$D = V = \bar{V} + F_{2V} = 64.6$ kmol/h $+ 0.8F_2 = 133.6$ kmol/h

$D = 133.6$ kmol/h

$V = 133.6$ kmol/h

⇒

CHECK OVERALL BAL.

$F_1 + F_2 = D + B$

$100 + 80 = 133.6 + 46.4$ ✓

Now solving for y_D

$F_1 z_1 + F_2 z_2 = D y_D + B x_B$

$y_D = 0.675$

⇓

Finding op. Lines

TOP: $V_y + F_1 z_1 = Lx + D y_D$

$\ast\ y = \frac{Lx}{V} + \frac{D y_D - F_1 z_1}{V}$

BOTTOM: $\bar{V}_y + B x_B = \bar{L} x$

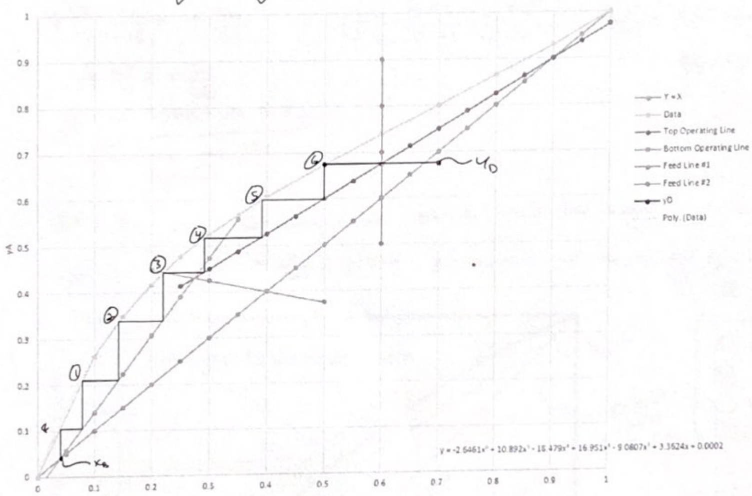
$\ast\ y = \frac{\bar{L}}{\bar{V}} x - \frac{B}{\bar{V}} x_B$

Graph OP Lines w/ data & Feed lines from above

F_1: $x = 0.6$

F_2: $y = -\frac{1}{4}x + \frac{1}{2}$

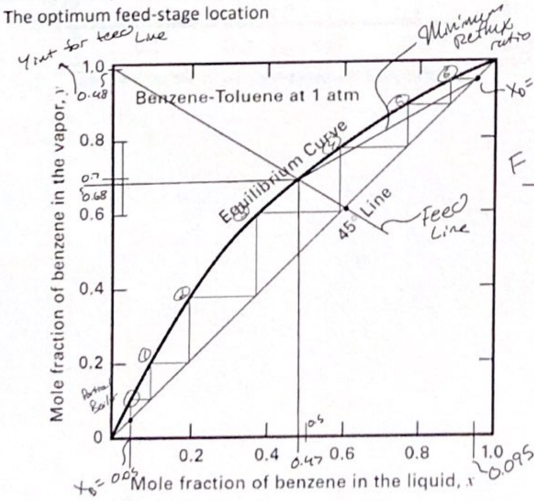
$D = 133.6$ kmol/h
$y_D = 0.675$
Optimum feed stage for feed #2 = STAGE #3
TOTAL # of STAGES = 6 + Reboiler

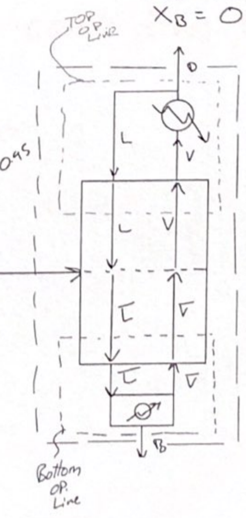


4) Four hundred and fifty lbmol/h of a mixture of 60 mol% benzene (LK) and 40 mol% toluene (HK) is to be separated into a liquid distillate and a liquid bottom product of 95 mol% and 5 mol% benzene, respectively. The feed enters the column with a molar percent vaporization equal to the distillate-to-feed ratio. Use McCabe-Thiele method to compute, at 1 atm (101.3 kPa):

(a) N_{min},
(b) R_{min},
(c) Number of equilibrium stages N, for $R/R_{min} = 1.3$,
(d) The optimum feed-stage location

$F = 450$ lbmol/h
$z = 0.60$ (Benzene)
$x_D = 0.95$
$x_B = 0.05$





$F = D + B$ (2 unknowns & 2 eq.)

$Fz_1 = Dx_D + Bx_B$

$D = F - B$

$Fz_1 = Fx_D - Bx_D + Bx_B$

$Fz_1 - Fx_D = B(x_B - x_D)$

$B = \frac{(Fz_1 - Fx_D)}{(x_B - x_D)}$

$B = \frac{450(0.60 - 0.95)}{(0.05 - 0.95)}$

$B = 175$ lbmol/h

$D = 275$ lbmol/h

Distillate to feed ratio $(D/F) = \frac{275}{450}$

Molar % vaporization $= \frac{11}{18} = 0.61$

* Feed Line *

$q = \frac{L}{F} = (1 - \frac{11}{18}) = \frac{7}{18}$ $z = 0.60$

$R = \frac{L}{D}$

$y = \frac{q}{q-1}x - \frac{z}{q-1}$

slope tells us: $\frac{L}{V} = \frac{L/D}{\frac{L+D}{D}} = \frac{L/D}{\frac{L}{D}+1} = \frac{R}{R+1}$ find slope & set equal to

$y = \frac{\frac{7}{18}}{\frac{7}{18}-1}x - \frac{\frac{6}{10}}{\frac{7}{18}-1}$

approx intersection of feed w/ data: (0.47, 0.68) w/ (0.95, 0.95)

$y = \frac{7}{7-18}x + \frac{54}{55}$

$\left(\frac{L}{V}\right)_{min} = \frac{0.95-0.68}{0.95-0.47} = \frac{0.27}{0.48} = \frac{27}{48}$ now using $\frac{L}{V} = \frac{R}{R+1}$

$\frac{27}{48}(R_{min}+1) = R_{min}$ $R_{min} = \frac{\frac{27}{48}}{(1-\frac{27}{48})} = \frac{27}{21} = 1.29$

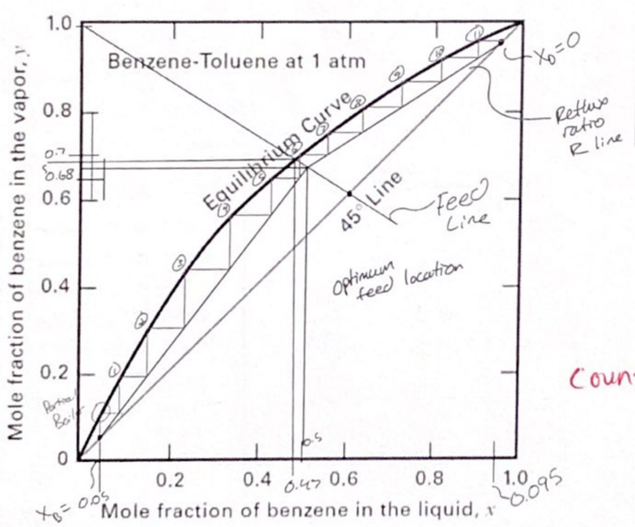
$y = \frac{-7}{11}x + \frac{54}{55}$

$\underbrace{y = -0.636x + 0.982}_{\text{plotting on the graph}}$

If $\frac{R}{R_{min}} = 1.3$ we can solve for R then for slope

$R = 1.67 \Rightarrow$ slope $= \frac{R}{R+1} = 0.626$ & intersects w/ $x_D, x_D \Rightarrow y = 0.626x + 0.356$

@ $x = 0.47$ $y = 0.65$



A) 6 STAGES & PARTIAL REBOILER
B) $R_{min} = 1.29$
C) 11 STAGES & PARTIAL REBOILER
D) OPTIMUM FEED LOCATION IS STAGE 6

$0.626x + 0.356 = \frac{-7}{11}x + \frac{54}{55}$

$x \simeq 0.50$ intersection

Count from top.